

Thi-Nhung Nguyen

PhD Researcher | Agentic AI & Foundation Models

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[Personal Website](#) | [Google Scholar](#) | [LinkedIn](#)

RESEARCH PROFILE

PhD researcher with 5+ years of experience spanning applied machine learning research and production AI. Builds and evaluates LLM and agentic systems, with research spanning multi-agent safety, autonomy, and robustness, alignment, long-horizon planning, fine-tuning, and automated evaluation. Published at EMNLP, COLM, CoNLL, NAACL, and LREC-COLING; previously led LLM and retrieval-augmented generation products from experimentation to deployment.

My research aims to advance agentic systems and frontier models for planning and reasoning, and to enable large-scale coordination through agentic collectives and economies. This includes intelligent orchestration and communication, self-improvement, and agentic safety and autonomy.

EDUCATION

Monash University

Doctor of Philosophy, Data Science & AI

Jun. 2025–Present
Melbourne, Australia

- Research: safe and autonomous multi-agent systems, planning, and LLM alignment.
- Supervisors: [Prof. Dinh Phung](#), [Prof. Thuy-Trang Vu](#), and [Dr. Linhao Luo](#).

Hanoi University of Science and Technology

Bachelor of Computer Science

Aug. 2017–Aug. 2021
Hanoi, Vietnam

- Thesis: *An Uncertainty-Aware Encoder for Aspect Detection*; Best Thesis Presentation Award.

TECHNICAL SKILLS

Programming & ML Frameworks: Python, PyTorch, Hugging Face Transformers, FastAPI, Rasa, LangChain, LangGraph

LLMs & Post-Training: Fine-tuning, alignment, reinforcement learning, automated data generation, model evaluation

RAG & Retrieval: RAG, hybrid search, Text-to-SQL, Text-to-Cypher

Agentic AI: AI agents, multi-agent systems, planning, workflow orchestration, safety and evaluation

Databases & Infrastructure: Elasticsearch, Qdrant, Neo4j, MongoDB, Docker

RESEARCH EXPERIENCE

PhD Researcher & Research Assistant

Monash University

Jun. 2025–Present
Melbourne, Australia

- Supervisors: [Prof. Dinh Phung](#), [Prof. Thuy-Trang Vu](#), and [Dr. Linhao Luo](#).
- Research safe and autonomous multi-agent systems, encompassing LLM alignment, long-horizon planning and reasoning, intelligent orchestration and communication, self-improvement, and safety evaluation.
- Developed research and evaluation methods for truth consistency, long-term planning, retrieval-augmented generation (RAG), table understanding, and emergent bias in multi-agent systems.
- Contribute to two research-assistant projects: multi-agent safety (Jan. 2026–Present) and LLM alignment (Apr.–Jun. 2026).
- Published three papers at EMNLP, COLM, and CoNLL, with additional work under review.

AI Research Resident

VinAI Research (now Qualcomm AI Research)

Mar. 2022–Apr. 2024
Hanoi, Vietnam

- Research supervisor: [Assoc. Prof. Thien Huu Nguyen](#) and [Dr. Dat Quoc Nguyen](#).
- Researched weakly supervised learning and information extraction, resulting in four papers at EMNLP, NAACL, and LREC-COLING.
- **PhoGPT**: contributed to an open-source generative pre-trained language model for Vietnamese.

Research Student

Data Science Laboratory, HUST

Oct. 2019–Mar. 2022
Hanoi, Vietnam

- Supervisor: [Dr. Kiem-Hieu Nguyen](#).
- Developed weakly supervised methods for aspect detection; published at EMNLP Findings 2021.

INDUSTRY EXPERIENCE

Senior AI Engineer, LLM Team Lead <i>MISA JSC</i>	Dec. 2024–May 2025 <i>Hanoi, Vietnam</i>
• Supervisor: Dr. Trung Bui (Adobe Research). • Led fine-tuning and evaluation of Vietnamese LLMs, reaching #1 on the VMLU leaderboard as of Mar. 25, 2025 and a top-five position in a national government-organized evaluation. • Built automated data-generation and evaluation pipelines, fine-tuned embedding models, local LLMs, and domain agents for accounting and financial analysis. • Directed delivery across LLM alignment, safety alignment, RAG, and multi-agent projects while mentoring engineers and communicating with stakeholders.	
Senior AI Engineer, Technical Lead <i>MISA JSC – Legal Q&A System</i>	Apr. 2024–Dec. 2024 <i>Hanoi, Vietnam</i>
• Led a RAG-based legal assistant over 1,000+ documents (3–300 pages each), achieving > 95% search accuracy and > 90% answer accuracy on 1,000+ expert-curated questions. • Translated product requirements into retrieval and evaluation tasks using Python, Rasa, databases, chunking, RAG, and Text-to-SQL; product success helped secure two new projects.	

SELECTED PUBLICATIONS

TriAlign: Towards Universal Truth Consistency in Personalized LLM Alignment Thi-Nhung Nguyen , Linhao Luo, Rollin Omari, Junae Kim, Thuy-Trang Vu, Dinh Phung <i>EMNLP 2026</i>	
The Social Cost of Intelligence: Emergence, Propagation, and Amplification of Stereotypical Bias in Multi-Agent Systems Thi-Nhung Nguyen , Linhao Luo, Amardeep Kaur, Rollin Omari, Tamas Abraham, Junae Kim, Thuy-Trang Vu, Dinh Phung <i>Preprint, 2026</i>	
Improving Table Understanding with LLMs and Entity-Oriented Search Thi-Nhung Nguyen , Hoang Ngo, Dinh Phung, Thuy-Trang Vu, Dat Quoc Nguyen <i>COLM 2025</i>	
Planning for Success: Exploring LLM Long-term Planning Capabilities in Table Understanding Thi-Nhung Nguyen , Hoang Ngo, Dinh Phung, Thuy-Trang Vu, Dat Quoc Nguyen <i>CoNLL 2025</i>	
BKEE: Pioneering Event Extraction in the Vietnamese Language Thi-Nhung Nguyen , Bang Tran, Trong-Nghia Luu, Kiem-Hieu Nguyen, Thien Huu Nguyen <i>LREC-COLING 2024</i>	
A Self-Enhancement Multitask for Unsupervised Aspect Category Detection Thi-Nhung Nguyen , Hoang Ngo, Kiem-Hieu Nguyen, Tuan-Dung Cao <i>EMNLP 2023</i>	
An Uncertainty-Aware Encoder for Aspect Detection Thi-Nhung Nguyen , Kiem-Hieu Nguyen, Young-In Song, Tuan-Dung Cao <i>Findings of EMNLP 2021</i>	

TEACHING

Teaching Assistant – Natural Language Processing, Monash University	Semester 1, 2026
Teaching Assistant – Natural Language Processing, VinUniversity	2024
Teaching Assistant – Web Mining and Graph Analytics, HUST	2021

SELECTED HONORS

Best Thesis Presentation Award, HUST, 2021
Microsoft YouthSpark Scholarship, 2017 and 2018